



HR Overtime Management System

Complete Technical, Configuration & Test Documentation — Odoo 19

Module: hr_overtime_management (+ optional hr_overtime_payroll)

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Database: odoo19 | Config: debian/odoo.conf



Table of Contents

- 1. Executive Summary
- 2. What Was Customized vs Standard Odoo
- 3. Business Objectives
- 4. Delivered Modules & Dependencies
- 5. System Architecture & Workflow
- 6. Database Models
- 7. Cost Calculation
- 8. Security, Groups & Multi-Company
- 9. User Interface & Menus
- 10. Configuration Guide (Step by Step)
- 11. Server & Database Setup (odoo.conf)
- 12. End-User Guide
- 13. Automated Test Cases
- 14. Issues Fixed During Delivery
- 15. Technical File Structure
- 16. Upgrade & Maintenance



1. Executive Summary

This document describes the complete HR Overtime Management solution built on Odoo 19 for your organization. Employees submit overtime linked to projects and tasks; requests flow through department manager → optional upper manager → HR approval; costs are computed from contract wage and configurable rate multipliers (Regular, Weekend, Day Off); approved overtime can create timesheet lines and optionally feed payroll.

The solution follows Odoo best practices: reusable approval engine (similar to Time Off), per-company configuration, no duplicate payroll tables, scoped sudo() for multi-company reads (not blanket security bypass), and friendly UI snackbars for access errors.

2. What Was Customized vs Standard Odoo

2.1 Reused from Standard Odoo

Standard Odoo	How We Use It
hr.employee + parent_id	Manager / upper-manager approval chain
hr.version (contract wage)	Hourly cost from employee wage
project.project / project.task	Project & task on each request
account.analytic.line	Optional timesheet on HR approval
resource.calendar.leaves	Public holidays for Day Off type
mail.thread / mail.activity	Chatter and approval reminders
res.company + company_ids	Multi-company branches and record rules
ir.attachment	Supporting documents on requests

2.2 Custom-Built Components

Component	Purpose
hr.overtime.request	Main overtime record and workflow
hr.overtime.type	Regular / Weekend / Day Off rate categories per branch
hr.overtime.approval.line	Audit trail per approval step
hr.approval.chain.service	Resolves manager → HR chain from employee hierarchy
hr.approval.chain.mixin	Submit, approve, refuse workflow
Auto overtime type	Type picked from date — employee cannot choose manually
Per-company type provisioning	Each branch gets 3 types on company create
overtime_error_handler.js	Snackbar for company/access errors on overtime forms
Refuse wizard	Mandatory reason when refusing
QWeb PDF report	Printable approval document
hr_overtime_payroll (optional)	Glue to paystub inputs

2.3 Key Customizations Applied



- Datetime fields (start_datetime / end_datetime) instead of separate date + float time.
- Overtime Type is computed and read-only for employees.
- Weekend = Friday & Saturday by default (overtime_weekend_weekdays = 4,5).
- Each company branch has independent rate multipliers.
- total_cost = hours × hourly_cost × rate_multiplier.
- Project domain limited to employee company + shared projects.
- Approval lines created via workflow sudo() — employees do not need direct ACL on lines.
- Managers see requests via approval_line_ids OR current_approver_id record rules.
- company_label Char field avoids direct res.company reads in the UI.

3. Business Objectives

- Employees submit overtime with project, task, and attachments.
- Multi-level approval mirrors organizational hierarchy + mandatory HR.
- Pay rates configurable per branch without code changes.
- Approved overtime can post to timesheets automatically.
- Cost derived from contract wage, not a parallel engine.
- Reusable approval chain for future HR request types.
- Multi-company safe: employee company drives record scope.

4. Delivered Modules & Dependencies

Module	Required	Purpose	Depends On
hr_overtime_management	Yes	Core overtime app	hr, hr_timesheet, project, mail, resource
hr_overtime_payroll	No	Payslip overtime input	hr_overtime_management, hr_payroll

5. System Architecture & Workflow

5.1 Approval Chain

- Step 1: employee.parent_id (or dept manager) → dept_manager
- Step 2: parent of dept manager (if different) → upper_manager



- Step 3: user in group_overtime_hr_officer → hr (always final)

Two-stage chain: dept_manager → HR. Three-stage: dept_manager → upper_manager → HR.

5.2 Workflow States

State	Meaning	Who Can Act
draft	Employee editing	Employee (owner)
submitted	Waiting first approver	Dept manager
manager_approved	Dept approved	Upper manager or HR
upper_manager_approved	Upper approved	HR officer
hr_approved	Fully approved	Read-only; timesheet created
refused	Rejected	Employee may reset to draft
cancel	Cancelled by employee	Employee

5.3 Automatic Overtime Type

Category	When	Default Multiplier
regular	Sun–Thu (configurable working days)	1.5×
weekend	Fri & Sat (weekdays 4,5)	2.0×
day_off	Public holiday on employee calendar	2.5×

If the period spans multiple categories, the highest multiplier wins.

6. Database Models

6.1 hr.overtime.request

Field	Description
name	Sequence OT/YYYY/MM/00001
employee_id	Employee who worked overtime
employee_company_id	Integer — employee branch (avoids company ACL on compute)
company_id	Stored FK — always employee's company
company_label	Display name via sudo (UI field)
start_datetime / end_datetime	Overtime period
overtime_hours	Computed duration
overtime_type_id	Auto-computed category
project_id / task_id	Timesheet project and task
state	Workflow state
current_approver_id	Active approver user
hourly_cost / total_cost	Computed monetary fields
analytic_line_id	Timesheet line after HR approval
attachment_ids	Supporting documents (company_id=False)



6.2 hr.overtime.type

Field	Description
category	regular weekend day_off
rate_multiplier	Pay multiplier per branch
company_id	Company branch (one set per company)

6.3 hr.overtime.approval.line

Field	Description
role	dept_manager upper_manager hr
approver_id	Assigned approver
state	pending to_approve approved refused
comment	Refusal reason

6.4 res.company Extensions

Setting	Default	Purpose
overtime_generate_analytic_line	True	Create timesheet on HR approval
overtime_default_type_id	Regular	Working day type
overtime_weekend_type_id	Weekend	Weekend type
overtime_holiday_type_id	Day Off	Holiday type
overtime_weekend_weekdays	4,5	Fri=4, Sat=5
overtime_daily_hours_cap	4.0	Warning threshold
overtime_hours_per_month	173.33	Wage ÷ hours fallback

7. Cost Calculation

Formula: $\text{total_cost} = \text{overtime_hours} \times \text{hourly_cost} \times \text{rate_multiplier}$

- Primary hourly: from hr.version wage normalized by work schedule
- Fallback: $\text{wage} \div \text{overtime_hours_per_month}$ (default 173.33)

Example: 2 h Regular at 1.5× with \$45.81/h → total ≈ \$137.43

8. Security, Groups & Multi-Company

8.1 Security Groups

Group	Users	Access
group_overtime_user	All internal users	Create/read own requests; edit draft
Managers (record rules)	Assigned approvers	Read/write pending requests



group_overtime_hr_officer	HR team	Read all; final HR approval
group_overtime_admin	Admins	Full config access

8.2 Record Rules (hr.overtime.request)

- Employee read own: employee_id.user_id = current user
- Employee write draft / cancel / reset: state-based domains
- Manager: current_approver_id = user OR active approval_line approver
- Multi-company: company_id in company_ids
- HR officer: read all; write at HR approval stage

8.3 Multi-Company Best Practice (Implemented)

Odoo's built-in company rule employee restricts res.company reads to the user's Allowed Companies. The overtime module handles this correctly:

- Overtime company_id always equals employee's company — never the session company.
- Project dropdown limited to employee company + shared (no company) projects.
- Validation raises UserError if project branch not in Allowed Companies.
- Scoped sudo() for company settings reads (_sudo_company) and web_read for approvers.
- Attachments stored with company_id=False to avoid cross-company errors.
- Overtime menus set allowed_company_ids to user.company_ids.
- Snackbar (not security bypass) when company AccessError occurs on overtime forms.
- Admin fix: add branch under Settings → Users → Allowed Companies if cross-branch needed.

8.4 Approval Line ACL

Employees have read-only ACL on hr.overtime.approval.line. Create/write/unlink happens inside the workflow engine (hr.approval.chain.mixin) using scoped sudo() after permission checks (_can_user_approve_line, ownership).

9. User Interface & Menus

Menu	Who	Purpose
Overtime → My Requests	Employees	Own overtime list and form
Overtime → To Approve	Managers / HR	Pending items assigned to you
Overtime → All Requests	HR officers	Full list across company
Overtime → Configuration → Overtime Types	Admins	Rate multipliers per branch
Settings → Employees → Overtime	Admins	Company overtime settings



Views: List, Form (statusbar, approval tab, chatter), Kanban, Search, Pivot, Graph, PDF report.

10. Configuration Guide (Step by Step)

10.1 Initial Installation

- 1. Ensure Odoo 19 is running with debian/odoo.conf (see Section 11).
- 2. Database name must be odoo19 (not odoo).
- 3. Apps → Update Apps List → search HR Overtime Management → Install.
- 4. Optional: install hr_overtime_payroll if payroll integration is required.
- 5. Upgrade after any code change: -u hr_overtime_management --stop-after-init

10.2 Per Company Branch

- 1. Switch to the company branch (top-bar company selector).
- 2. Verify 3 overtime types exist (auto-created): Regular, Weekend, Day Off.
- 3. Overtime → Configuration → Overtime Types → edit Rate Multiplier per type.
- 4. Settings → Employees → Overtime → set weekend weekdays (default 4,5 = Fri/Sat).
- 5. Set Daily Hours Warning Cap and Hourly Cost Divisor if needed.
- 6. Enable/disable Timesheet Integration (overtime_generate_analytic_line).

10.3 Users & HR Officers

- 1. Settings → Users → assign Officer: Overtime HR Approval to HR staff per branch.
- 2. Each employee must have an hr.employee linked to their user.
- 3. Set Manager (parent_id) on employee for approval chain resolution.
- 4. Allowed Companies: add every branch the user must read (e.g. SF + Chicago if cross-branch).
- 5. Default Company should match the employee's company.

10.4 Projects & Employees

- 1. Projects used for overtime must have Allow Timesheets enabled.
- 2. Employee sees only projects from their company or shared projects.
- 3. Ensure hr.version has wage set for cost calculation.
- 4. Public holidays: Resource → Working Times → Time Off (calendar leaves).



10.5 Test Users (odoo19 database)

User	Role	Notes
john_doe (id=5)	Employee	San Francisco company; submits overtime
toto@gmail.com (id=6)	Dept manager	Manager of john_doe; approves/refuses
admin	HR / Admin	HR approval and configuration

11. Server & Database Setup (odoo.conf)

Configuration file: debian/odoo.conf

Setting	Value	Notes
addons_path	addons,extra_addons,odoo/addons	Includes custom modules
db_host	127.0.0.1	PostgreSQL host
db_port	5432	PostgreSQL port
db_user	odoo	Database user
http_port	8069	Web interface URL
Database	odoo19	Target database for this project

11.1 Start & Upgrade Commands

Start server:

```
python odoo-bin -c debian\odoo.conf -d odoo19
```

Upgrade module after code changes:

```
python odoo-bin -c debian\odoo.conf -d odoo19 -u hr_overtime_management --stop-after-init
```

Run automated tests:

```
python odoo-bin -c debian\odoo.conf -d odoo19 -u hr_overtime_management --test-tags hr_overtime --stop-after-init
```

Regenerate this documentation:

```
python extra_addons/hr_overtime_management/doc/generate_full_documentation.py
```

12. End-User Guide

12.1 Employee — Submit Overtime

- Overtime → My Requests → New.
- Enter start/end datetime, project, task, description.



- Overtime Type and cost compute automatically.
- Attach files if needed → Save → Submit.

12.2 Manager — Approve or Refuse

- Overtime → To Approve (only items assigned to you).
- Open request → Approve or Refuse.
- Refuse requires a reason (wizard).
- After refusing, list is empty until a new request is assigned to you.

12.3 HR — Final Approval & Config

- Users in Officer: Overtime HR Approval perform final approval.
- Overtime → All Requests for full visibility.
- Configure rates under Overtime → Configuration → Overtime Types.

12.4 Company Access Snackbar

If you see a warning snackbar about Allowed Companies, either pick a project from your own branch or ask an administrator to add the missing company under your user profile.

13. Automated Test Cases

File: tests/test_hr_overtime.py | Tag: hr_overtime | Run with --test-tags hr_overtime

Test Method	What It Verifies
test_overtime_hours_same_day	Duration 18:00–21:30 = 3.5 hours
test_overtime_hours_overnight	Overnight span = 4.0 hours
test_chain_single_manager_two_stages	Solo employee: dept_manager → HR
test_chain_manager_upper_manager_three_stages	Full 3-step chain
test_chain_keeps_hr_when_same_user_as_manager	HR stage never removed
test_refusal_at_stage_one_stops_chain	Refuse stops workflow; later lines stay pending
test_non_approver_cannot_approve	Intruder gets AccessError
test_internal_user_has_overtime_group	base.group_user implies overtime user
test_employee_can_create_draft_without_approve	Employee creates own draft
test_employee_cannot_approve_submitted_request	Employee cannot self-approve
test_cross_company_project_requires_allowed_company	UserError without branch; OK after Allowed Companies
test_employee_sees_overtime_menu	Overtime menu visible to employees
test_auto_overtime_type_weekday_regular	Sunday → Regular type
test_auto_overtime_type_friday_weekend	Friday → Weekend type
test_auto_overtime_type_saturday_weekend	Saturday → Weekend type
test_auto_overtime_type_public_holiday	Calendar leave → Day Off type
test_cost_computation_all_types	total_cost for Regular, Weekend, Holiday multipliers
test_full_approval_flow_three_stages	Dept → Upper → HR → hr_approved



test_full_approval_flow_two_stages	Solo manager → HR → hr_approved
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13.1 Manual Test Checklist

#	Scenario	Expected Result
1	Employee creates draft overtime	Saved; type auto-set; company = employee branch
2	Employee submits	State submitted; approval lines created; activity to manager
3	Manager approves	State advances; next approver gets activity
4	Manager refuses with reason	State refused; snackbar not raw AccessError
5	HR final approval	hr_approved; timesheet line if enabled
6	Wrong company project	UserError or snackbar; not cryptic ACL dialog
7	Multi-company user john_doe	My Requests shows own records only
8	Manager toto To Approve	Shows only requests assigned to toto
9	PDF print	Report generates for approved request
10	Module upgrade	No missing column errors

14. Issues Fixed During Delivery

Issue	Cause	Resolution
res.company Chicago AccessError	Session company ≠ employee Allowed Companies	employee_company_id, allowed_company_ids, scoped sudo, snackbar
Cannot create approval.line	Employees lack create ACL on lines	Workflow sudo() in approval mixin
Manager cannot refuse (read error)	Record rules + wizard read overtime	Approver rule via approval_line_ids; wizard sudo
My Approvals empty	Filter used only current_approver_id	Domain + rule use approval lines; recompute approver
Blanket security bypass reverted	Earlier bypassed res.company rules	Proper validation + scoped sudo only
Submit crash all_user_ids	Odoo 19 API change	get_hr_responsible uses all_user_ids
Hourly cost \$0	hr.version not hr.contract	Read wage from version_id
Missing DB columns	Module not upgraded	SQL migration + -u hr_overtime_management
Menu server action TypeError	Context string not dict	literal_eval in merge
Archived overtime types	active=False types skipped	_ensure_overtime_types reactivates
web_read recursion	Missing super() in override	Fixed then replaced with scoped sudo
Attachment company error	Wrong company_id on files	company_id=False for overtime attachments

15. Technical File Structure



extra_addons/hr_overtime_management/

- models/hr_overtime_request.py — main logic, workflow, multi-company
- models/hr_approval_mixin.py — submit/approve/refuse engine
- models/hr_approval_chain_service.py — chain resolver
- models/hr_overtime_type.py — rate categories
- models/hr_overtime_approval_line.py — approval audit
- models/res_company.py — per-branch settings
- models/ir_attachment.py — attachment company fix
- security/security.xml — groups and record rules
- security/security_overtime_approver_access.xml — approver rule upgrade hook
- static/src/overtime_error_handler.js — snackbar handler
- wizard/hr_overtime_refuse_wizard.py — refusal dialog
- tests/test_hr_overtime.py — 19 automated tests
- hooks.py — post_init: provision overtime types
- doc/generate_full_documentation.py — this document generator

16. Upgrade & Maintenance

Current module version: 19.0.1.5.3

- After any Python/XML change: upgrade hr_overtime_management on odoo19.
- Hard-refresh browser (Ctrl+Shift+R) after JS changes.
- Use odoo19 database — not odoo.
- Do not delete company rule employee — fix Allowed Companies or use scoped sudo.
- Regenerate documentation: python doc/generate_full_documentation.py

— End of Document —